INCH-POUND
MIL-PRF-10509/3K
22 August 2023
SUPERSEDING
MIL-R-10509/3J
w/Amendment 4
12 June 2017

PERFORMANCE SPECIFICATION

RESISTOR, FIXED, FILM (HIGH STABILITY), STYLE RN70

INACTIVE FOR NEW DESIGN AFTER 9 MAY 1972.
USE MIL-PRF-55182/6.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-10509.

1. SCOPE

1.1 Scope. This specification covers the requirements for style RN70, high stability, film, fixed, resistors. This style is available in characteristics C, D, E, F and G.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification is identified by a PIN which consist of the style of this specification and a coded number. The PIN is in following form:

RN70	D1003F
Specification style number	Coded number

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information are as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A
FSC 5905

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DEPARTMENT OF DEFENSE SPECIFICATIONS

* MIL-PRF-10509 - Resistor, Fixed, Film (High Stability), General Specification for

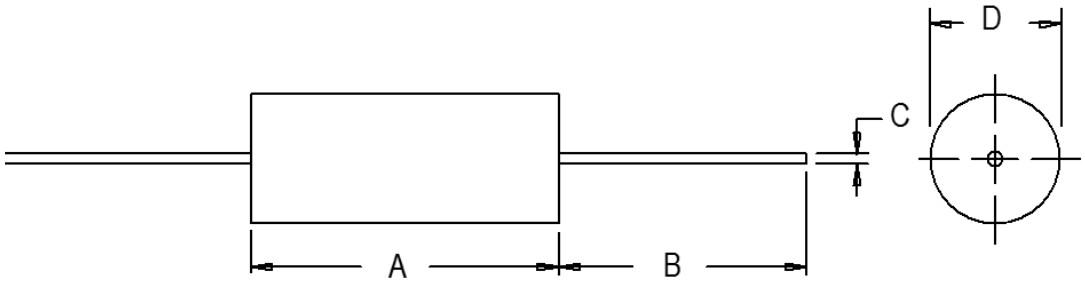
* (Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-10509.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1.



Dimension	Inches			mm		
A	0.725,	+0.125,	-0.250	19.05,	+3.18,	-6.35
B	1.500,	±0.125		38.10,	±3.18	
C	0.032,	±0.002		0.81,	±0.05	
D	0.250,	+0.125,	-0.090	6.35,	+3.18,	-2.29

NOTE:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 150 percent of the normal lead diameter, or 300 percent of the nominal lead diameter. The same drill size shall be used for each end of the body.
4. The leads shall be solderable to within 0.062 (1.58 mm) of the resistor body.

FIGURE 1. Style RN70.

3.3 Power rating. The power rating shall be 0.5 watt. For characteristic D and F, the power rating shall be 0.75 watt. Characteristics D and F were formerly rated at 1 watt; however, the 3/4 watt is a direct replacement for the 1 watt unit.

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3.4 Voltage rating. The maximum continuous working voltage shall not exceed 350 volts. For characteristic D and F, the maximum continuous working voltage shall not exceed 500 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows in table I.

TABLE I. Minimum and maximum resistance.

Characteristic	Resistance	
	Minimum (in Ohms)	Maximum (in Megohms)
D, G	10.0	2.49
C, E, F	24.9	1.0

3.6 Pure tin. The use of pure tin, as an underplate or final finish is prohibited both internally and externally. Tin content of resistor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.4).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-10509.

4.2 Terminal strength. Direct load shall be applied gradually until the load reaches 5 pounds.

4.3 Dielectric withstanding voltage. The magnitude of test voltage shall be 900 volts for atmospheric pressure and 300 volts for barometric pressure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The notes specified in MIL-PRF-10509 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this date of this specification, the applicable associated specification, and the complete PIN.

b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.

c. Packaging requirements (see 5.1).

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6.3 Substitution data. The resistors specified herein are not for use in design after 9 May 1972. They are authorized for use in design contracts effective prior to or on 9 May 1972. Existing equipment shall be supported by substitute resistors in MIL-PRF-55182/6 of the same resistance value, resistance tolerance, and characteristic performance of this specification regardless of failure rate designation. Resistors specified in MIL-PRF-55182/6 are preferred for design and regardless of failure rate designation can be used as substitutes for inactivated resistors of the same resistance value, tolerance, and characteristic performance of this specification. Characteristics C, F, and G were previously inactivated for design by revision F of this specification, dated 22 November 1968. Characteristic substitution shall be made as indicated in table II.

* TABLE II. Characteristic substitution. 1/

MIL-PRF-10509 Characteristic	Substitute MIL-PRF-55182 characteristic
C	H
D	K
E	J
G	C

1/ Characteristic F was removed from this table due to no direct replacement in MIL-PRF-55182/6.

6.4 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.5 Pulse applications. Designers are CAUTIONED on using these resistors in high power pulse applications. Since they have not been qualified nor tested for such applications, damage and premature failure are possible

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Navy - EC
Air Force - 85
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(Project 5905-2023-013)

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